RAJALAKSHMI ENGINEERING COLLEGE

Keerthivasan A 251001114-2025-IT

S2

Started on	Friday, 25 September 2026, 2:32 PM
State	Finished
Completed on	Friday, 25 September 2026, 2:36 PM
Time taken	4 mins 39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, i, j, max = 1;
    int a[100], dp[100];

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++)
        dp[i] = 1;

    for (i = 1; i < n; i++)
    {
        for (j = 0; j < i; j++)
        {
            if (a[j] <= a[i] && dp[j] + 1 > dp[i])
                dp[i] = dp[j] + 1;
        }
    }

    for (i = 0; i < n; i++)
    {
        if (dp[i] > max)
            max = dp[i];
    }

    printf("%d", max);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course